

Analysing and Tracking the Sentiment and Topics on Social Media

Social Media and Network Analysis

Wan Yi Lam

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Introduction

Social media is the giant treasure trove containing massive information which is collectively created by millions of individuals. With the development of text analysis and Natural Language Processing techniques, social media analysis could go further to discover the sentiment and topics in text. In this report, we will analyse the submissions about digital nomad in Reddit. Digital nomad is a new lifestyle that has become more popular since Covid-19 pandemic. This report provided the insights of the trending concepts and topics, and also the perceptions and feelings associated with digital nomad. We could discover new business opportunities from this market research in this booming lifestyle across the globe.

Data Collection

Firstly, the Reddit API PRAW is employed to gather latest submissions and comments from the digital nomad subreddit. The total number of submissions, which consists of 986 posts and 22,372 comments, was collected between 23rd February 2025 and 11th April 2025. All analyses in this report are based on this dataset. The whole text corpus from this dataset is mainly constructed by three components, including title and content of submission by the author, and also comments from other users.

In addition, the created timestamp of each submission and comment is recorded. In Reddit submission, the score records the number of upvotes for the submission that represents the submission is recognised by other users.

Summary Statistics

A series of summary statistics are conducted for the purpose of evaluating the data contents.

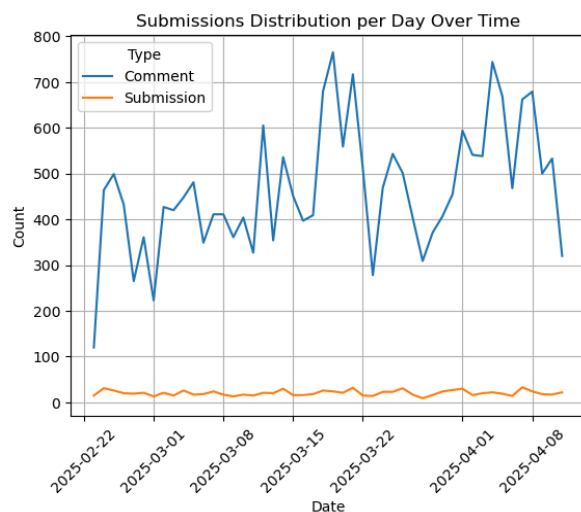


Figure 1: Submissions Distribution per Day Over Time

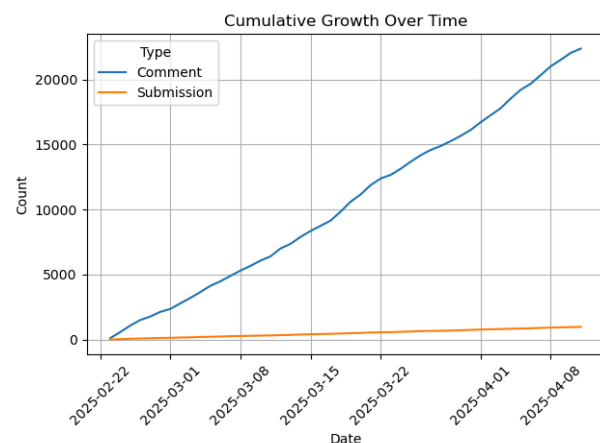


Figure 2: Submissions Distribution Over Time

As illustrated in Figure 1, the submission distribution per day over time indicates an average of approximately 20 new submissions per day. Despite the relatively low number of new submissions, the number of new comments is relatively high, ranging from 300 to 700 per day. This observation signifies the elevated level of user engagement in March 2025. As demonstrated in Figure 2, the cumulative growth of comments exhibits a rapid, accelerating trend, while the cumulative growth of submissions remains stable and low. This observation suggests that the users are actively engaged in discussing existing content, thereby underscoring the popularity of digital nomads on Reddit.

```

--- Top 5 Submissions by Score ---
Rank 1:
  Score: 1266
  Date:  2025-03-08 07:15:30
  Title: UAE gets 0 stars from me

Rank 2:
  Score: 1063
  Date:  2025-03-29 03:18:55
  Title: Vietnam probably has the best value for your money in the world right
now

Rank 3:
  Score: 769
  Date:  2025-03-24 18:12:40
  Title: Built the tool I wish existed when I started traveling - 5,698
cities, monthly Airbnb data, 100% free

Rank 4:
  Score: 527
  Date:  2025-03-31 03:36:40
  Title: First Thoughts After Returning From 5 Months in South America

Rank 5:
  Score: 463
  Date:  2025-04-01 16:40:56
  Title: If you want to move abroad, find a remote job first

```

Table 1 - Top 5 Submissions by Score

As shown in Table 1, the titles encompass a variety of publications, including travel critiques, destination recommendations, tools for travellers, and advice for living abroad. This diversity of content suggests a broad range of interests among the target audience. Of particular note is the prevalence of discourse centring on feedback pertaining to destinations shared by digital

nomads. This discussion is noteworthy as it provides a valuable source of data for further analysis.

Data Pre-processing and Exploration

Data preprocessing is the vital component before proceeding text analyses in order to transform the word forms of human language into the format for machines to parse and understand natural language.

First of all, the submission title, contents and comments are extracted from the existing text documents in dictionary format. Natural language processing tasks are employed to transform the text corpus in readable form for the machine, to make sure further sentiment analysis and topic modeling could be applied properly. These text preprocessing tasks are including,

- Convert texts to lowercase
- Tokenize all sentences into individual words
- Strip whitespace from each token
- Stopwords and punctuation Removal
- Stemming - Porter stemmer
- Filter out URLs
- Filter out tokens containing any digits
- Filter out number includes fraction and decimals

Original text: 'title': 'Latin America vs Eastern Europe'

Processed text: ['latin', 'america', 'vs', 'eastern', 'europ']

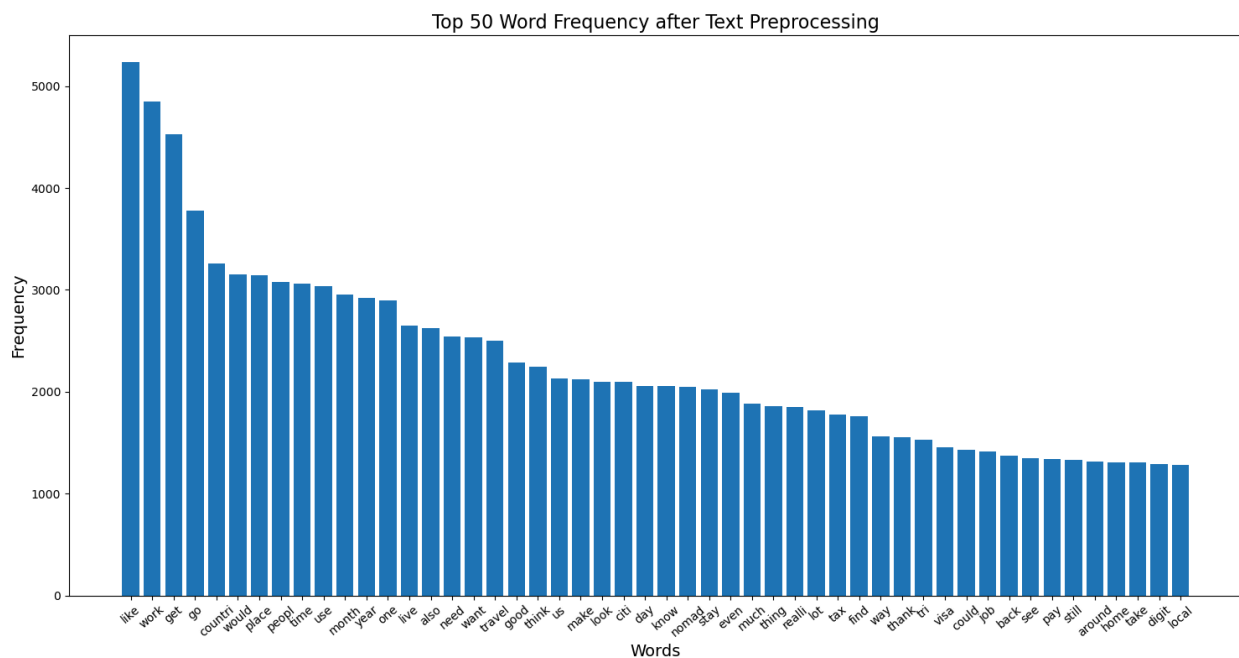


Figure 3: Top 50 Word Frequency after Text Preprocessing

Stemming is a process that can be used to group vocabulary in different word forms together, thus avoiding the word being counted separately. Consequently, a proportion of the words displayed in Figure 3 are represented by stems. Furthermore, the elimination of stopwords is an effective method of filtering out words of negligible significance. As illustrated in Figure 3, the most prevalent words are "like", "work", "get" and "go", and also some positive words frequently appear, including "like", "good", "thank", which offer a preliminary preview of this text corpus.

Analysis Approach

Sentiment Analysis

Considering this text corpus contains a wide variety of topics in digital nomad subreddit, it is complicated to perform the supervised approaches to analyse the sentiment with suitable labeling. Therefore, the following analyses are demonstrated by two unsupervised methods, the Count approach and the Vader approach.

Count Approach

The Count approach determines each token whether it is positive or negative by comparing it to the positive and negative word dictionaries (positive-words.txt and negative-words.txt). Then it simply calculates the difference between the number of positive words and negative words for each tokenized word in the submission. Hence, the sentiment outcome of each submission can be positive (more than zero), neutral (equal to zero) and negative (less than zero).

This approach is flexible and easy for implementation. However, the major limitation is that it only counts the word independently which means it cannot measure the relationship of word phrases. The positive and negative word dictionaries have been extracted from online movie reviews, which are gathered by Jeffrey Chan in 2023.

Example of positive outcome:

```
done, local, cab, tour, good, bad, place, latin, america, feel, like, could,
good, way, check, vibe, citi, know, would, trust, random, cab, tour, : 3
```

Vader Approach

The Vader approach is from the NLTK package for sentiment analysis using the Vader lexicon. This approach uses other lexical features, including: punctuation, capitalisation, degree modifier and presence of but, shifts polarity. These characteristics provide more feasible measurement for the degree of positiveness and negativeness, which is not limited to binary values. The outcomes of `polarity_scores()` contains negative, neural and positive sentiment values, all of these are normalised to the range of 0 to 1, with 1 being strongest. "Compound" value is the overall sentiment, between -1 (most negative) and 1 (most positive).

This approach was built from social media that will be the relevant approach to our analysis.

Example outcome:

```
yeah, total, zone, fact, second, se, bought, actual, anoth, gen, delet,  
comment, avoid, confus, anyon  
neg: 0.156, neu: 0.76, pos: 0.084, compound: -0.2263,
```

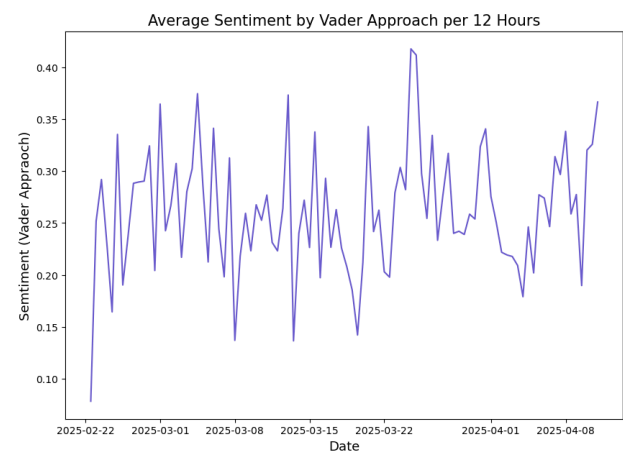
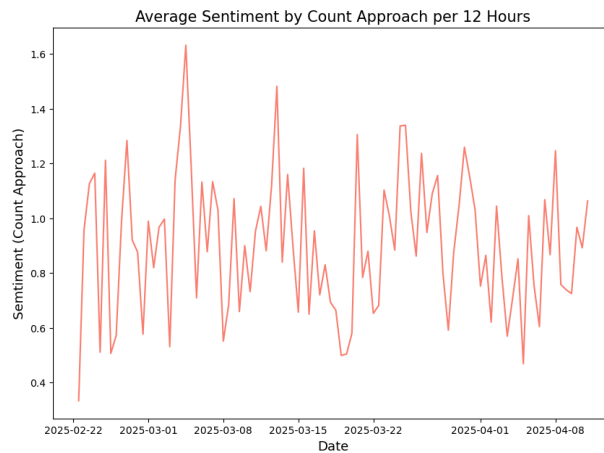


Figure 4: Average Sentiment by Count Approach per 12 Hours Figure 5: Average Sentiment by Vader Approach per 12 Hours
According to Figure 4 and 5, the average sentiment by different approaches indicates that the Vader approach is the most appropriate to reflect the level of sentiment score.

Topic Modeling

Latent Dirichlet Allocation (LDA) is a widely applied topic modelling technique which is the focus of this report. LDA can be regarded as a probabilistic model of text, whereby topics are generated through the identification of words and documents within these diverse topics according to probability distributions. The configuration of distinct parameters facilitates the modelling of documents pertaining to a variety of topics, from singleton topics to other topics through the co-occurrence of words. The objective of applying LDA is to identify the hidden topics in digital nomad by analysing large amounts of submissions and comments in this text corpus.

Analysis Insights

In this report, we are attempting to provide market insights of potential market segment digital nomad. Before answering the questions below, we also try to understand the activity times of potential customers on Reddit.

- What are the trending concepts and topics associated with digital nomad?
- What are the perceptions and feelings towards digital nomad?

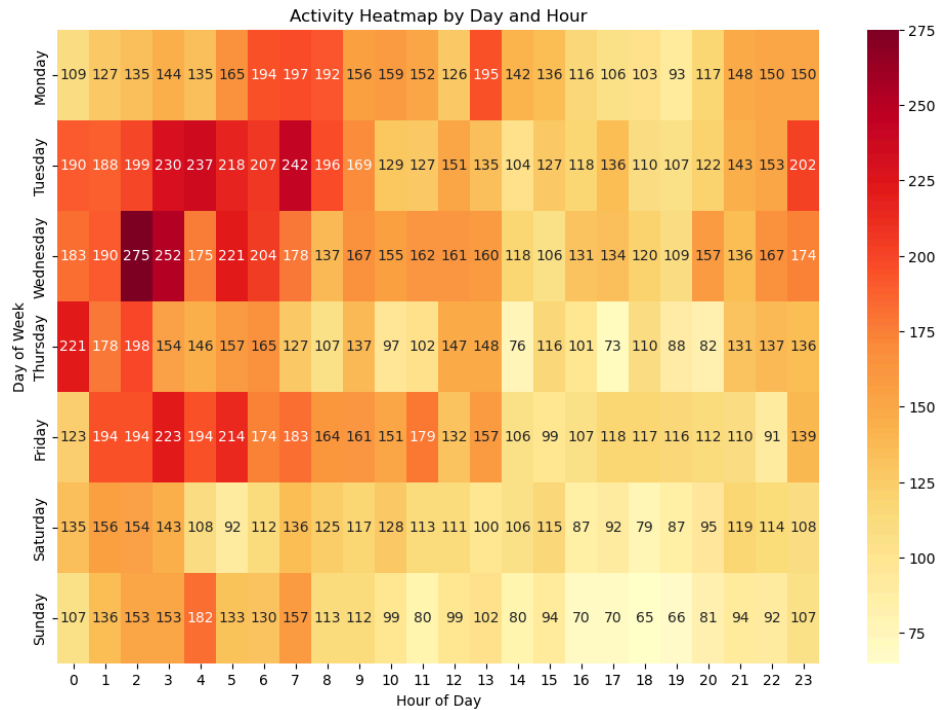


Figure 6: Activity Heatmap by Day and Hour

As shown in Figure 6, the hours of peak activity including posting submission and comments, for Reddit users occur between 11pm and 8am on Tuesday, Wednesday and Friday. This is the indication of the publishing time of the future marketing advertisements and promotions that maximise the reach rate of Reddit users.

Overall Sentiment Insights

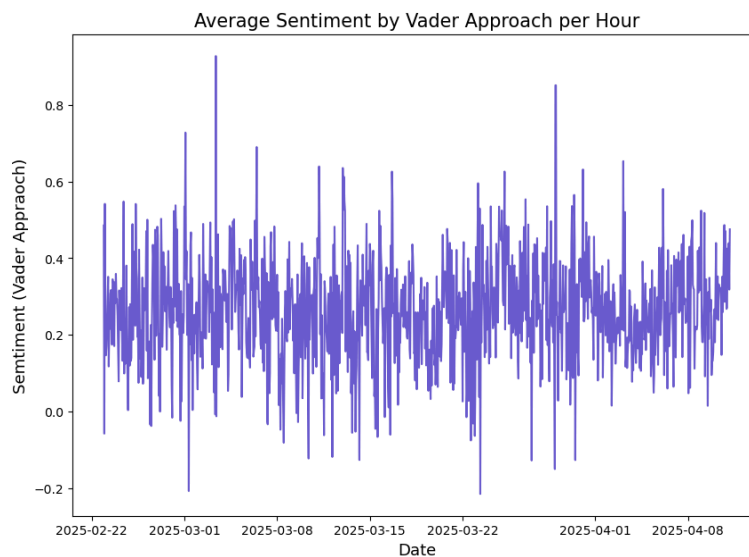


Figure 7: Average Sentiment by Vader Approach per Hour

According to Figure 7, the average sentiment score per hour conducted by the Vader method fluctuates within the range of -0.2 to 0.8 during specific intervals, with an average mean level ranging from 0.2 to 0.4 throughout the entire period of text analysis. Furthermore, the data in this text corpus was collected over a period of more than one month, from 23rd February 2025 to 11th April 2025, thereby ensuring the reliability and validity of the results. This suggests that, in general, users continue to hold favourable opinions of digital nomads.

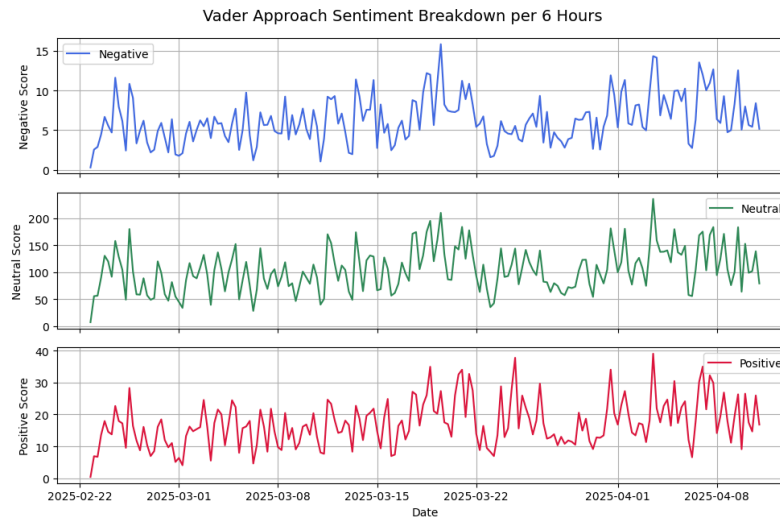


Figure 8: Vader Approach Sentiment Breakdown per 6 Hours

Referring to Figure 8, the trend of negative, neutral and positive sentiment demonstrates a consistent pattern over time. An increase in the total number of negative scores is observed, accompanied by a concomitant increase in the total number of neutral and positive scores. This finding indicates that the proportion of neutral scores increases over time. Reddit users engage in rational discussion on the platform.

Topic Modeling Insights

LDA parameters setting:

```
topicNum = 6
wordNumToDisplay = 15
featureNum = 2300
```

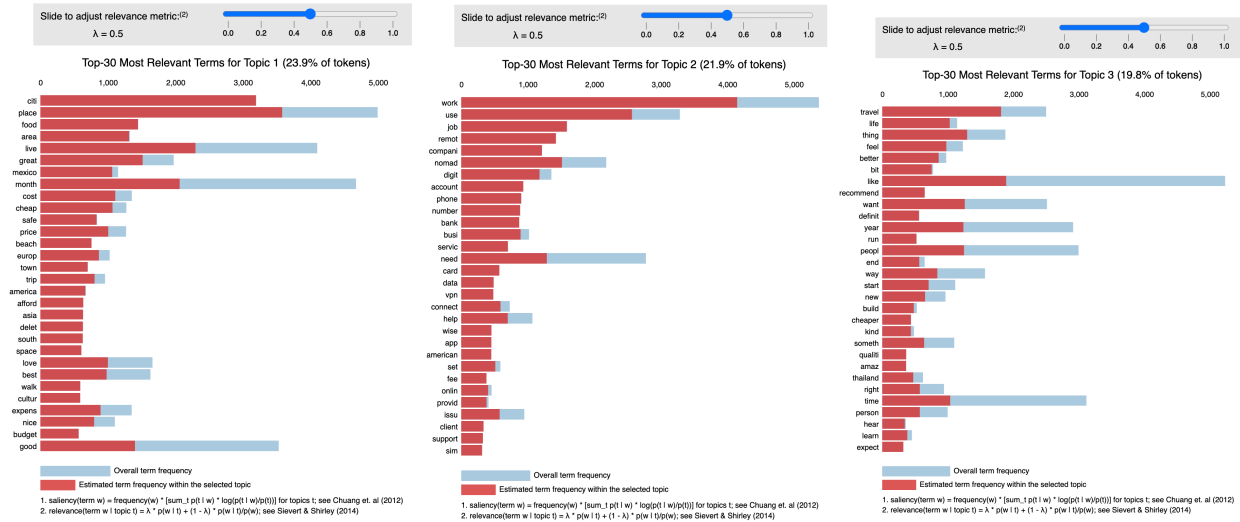



Figure 9: 6 Topics of Topic Modeling by LDA - examples of Topic 1(left), Topic 2(middle), Topic 3(right)

As a result of the LDA model, the Intertopic Distance Map (accessible via the output.html interface). This map meticulously displays six bubbles without overlap, indicating that each topic is distinct from the others. In addition, the application of a parameter, designated as λ , with a value of 0.5, is employed to regulate the co-occurrence of terms within different topics. Hence, the key terms listed on the bar charts are slightly different from the keys word of word clouds below.

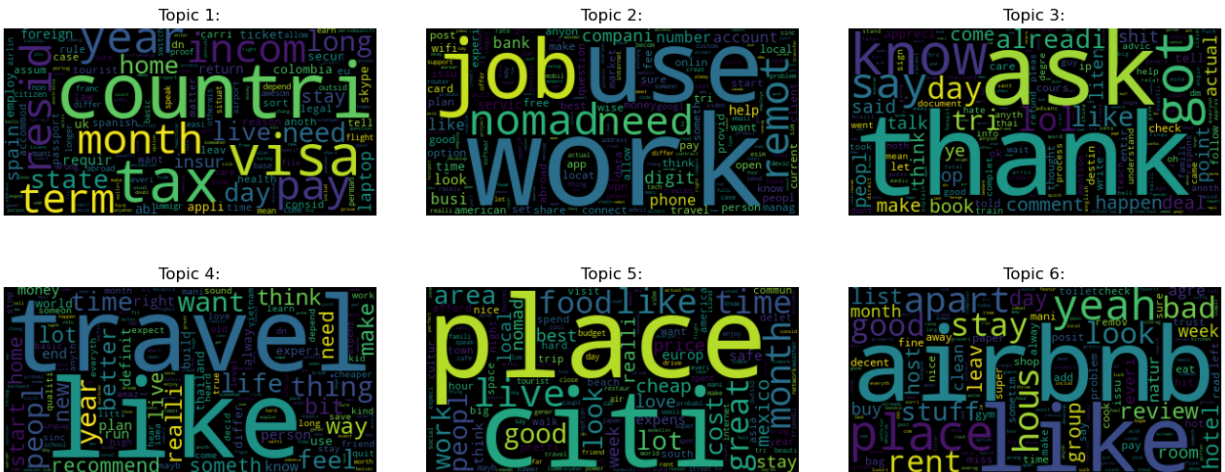


Figure 10: Word clouds - 6 Topics of Topic Modeling by LDA

Word clouds are a tool for visualising the frequency and probability of words by LDA model, whereby more frequently used and probable words appear larger in size.

A comparison of the visualisations of pyLDAvis and word clouds reveals the following hidden topics in digital nomad:

Topic 1 - Countries, tax, visa, residence, month, income - Destination regulation related

Topic 2 - Job, work, use, remote, need, bank, account, service - Employment and Service related

Topic 3 - Thank, ask, know, got, say - Enquiry related

Topic 4 - Travel, like, want, feel, recommend, think - Personal experience related

Topic 5 - Cities, place, food, live, food, great, cheap - Living experience related

Topic 6 - Airbnb, like, stay, rent, house, apartment - Housing related

These subjects offer invaluable insights, informing our understanding of the market needs that subsequently drive the development of new products and services. For instance, the provision of consultancy services in relation to visas and taxation, the promotion of foreign currency exchange by banking institutions. Other examples include mobile applications for social networking, the dissemination of information regarding employment opportunities and the short-term housing rental agency.

Conclusion

To conclude, different approaches of textual analysis of Reddit posts are carried out to find out the trending topics and feelings towards digital nomad. The present study focuses on the general level of activity exhibited by users on the Reddit platform. Subsequently, the positive sentiment pertaining to the aggregate submissions and comments was ascertained through the utilisation of Vader sentiment analysis. This finding indicates that the discourse within the subreddit has the potential to impart valuable insights and facilitate the exchange of knowledge concerning digital nomads. The LDA model is a valuable tool that can reveal hidden topics, which in turn can inform the development of new products and services.

References

Tran, K. (2025, April 7). *PYLDAVIS: Topic Modeling Exploration Tool Every NLP data scientist should know*. neptune.ai.<https://neptune.ai/blog/pyldavis-topic-modeling-exploration-tool>